

Experiment-05

Study of I-V Characteristics of Transistors (MOSFET & BJT) and Implementation of Logic Functions

CSE251 - Electronic Devices and Circuits Lab

Part-1 : MOSFET

Objective

1. To observe and understand the I-V characteristics of MOSFET.
2. To implement a Logic Function using MOSFETs and verify the truth tables.

Background Theory

Introduction to MOSFET

Three-terminal devices are far more useful than two-terminal ones (such as Diodes) because they can be used in a multitude of applications, ranging from signal amplification to digital logic and memory. The basic principle involved is the use of the voltage between two terminals to control the current flowing in the third terminal. In this way a three-terminal device can be used to realize a controlled source.

The control signal can be used to cause the current in the third terminal to change from zero to a large value, thus allowing the device to act as a switch. Switch is the basis for the realization of the logic inverter, which is a basic element of digital circuits.

There are two major types of three-terminal semiconductor devices: (i) MOSFET (Metal-Oxide Semiconductor Field-Effect Transistor) and (ii) BJT (Bipolar Junction Transistor). Although both of them offer unique features and areas of application, MOSFET has become by far the most widely used electronic device, especially in the design of integrated circuits (ICs).

There are two kinds of MOSFET: (i) NMOS and (ii) PMOS. In this experiment, we will study about the I-V characteristics of NMOS and design an NAND gate and a Boolean Logic Function using the NMOS transistor. The figure above shows the IC and the circuit diagram of a MOSFET (NMOS).

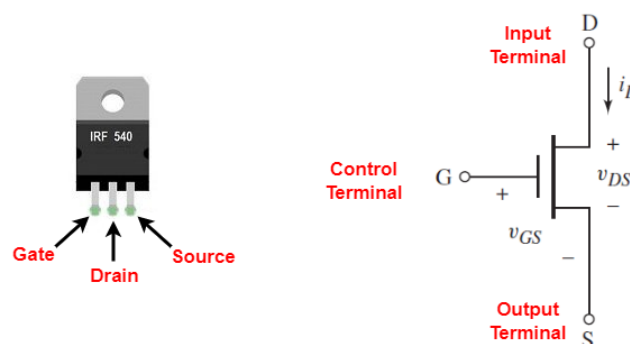


Figure 1: *NMOS IC (IRF540) and Circuit Diagram*

MOSFET as a Switch

MOSFET is a three-terminal device with a control terminal, an input terminal, and an output terminal. The control terminal of the MOSFET is called its gate G, the input terminal its drain D, and the output terminal its source S. MOSFET can act as a switch depending upon the applied voltage in the input terminal. A simple circuit model known as ‘Switch Model’ or ‘S-Model’ is used to model this behavior.

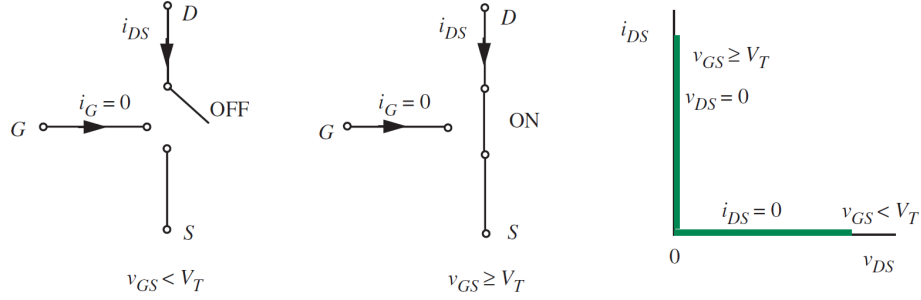


Figure 2: *S-Model of MOSFET*

Note that the current $I_G = 0$. We can summarize the S-model of the MOSFET in algebraic form as:

$$\text{when } \longrightarrow v_{GS} < V_T, \quad \text{then } \longrightarrow i_{DS} = 0 \quad \text{and} \quad \text{when } \longrightarrow v_{GS} \geq V_T, \quad \text{then } \longrightarrow v_{DS} = 0$$

MOSFET Switch Implementation of Logic Gates and Logic Functions

Switches can be used to build logic gates. As we can use MOSFET as a switch, logic gates can be built using MOSFETs. The following subsections discuss two logic gates, NOT gate and NAND gate, which are implemented using MOSFETs. We can also implement various Logic Functions using MOSFETs. To demonstrate this thing, a logic function will be implemented in Task-01.

NOT Gate/Inverter Using MOSFET

Consider the circuit shown in following figure, which comprises a MOSFET and a load resistor powered by a supply voltage V_S . This circuit acts as a NOT Gate/Inverter.

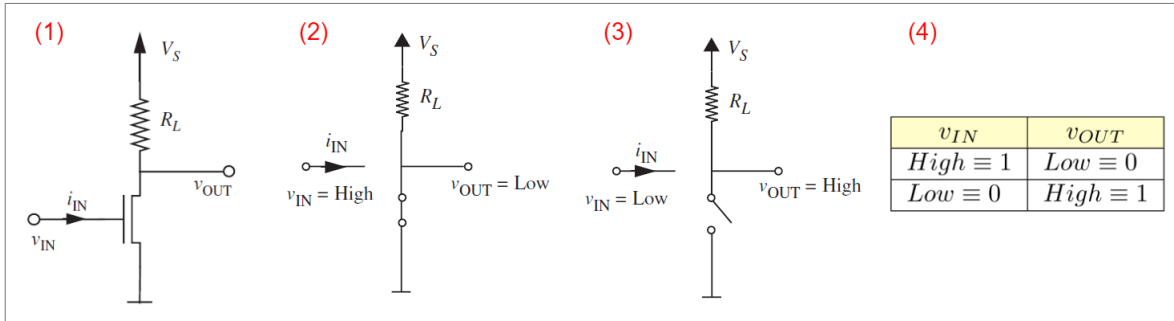


Figure 3: (1) *MOSFET NOT Gate* (2) *When Input is '1'* (3) *When Input is '0'* (4) *Truth Table*

NAND Gate Using MOSFET

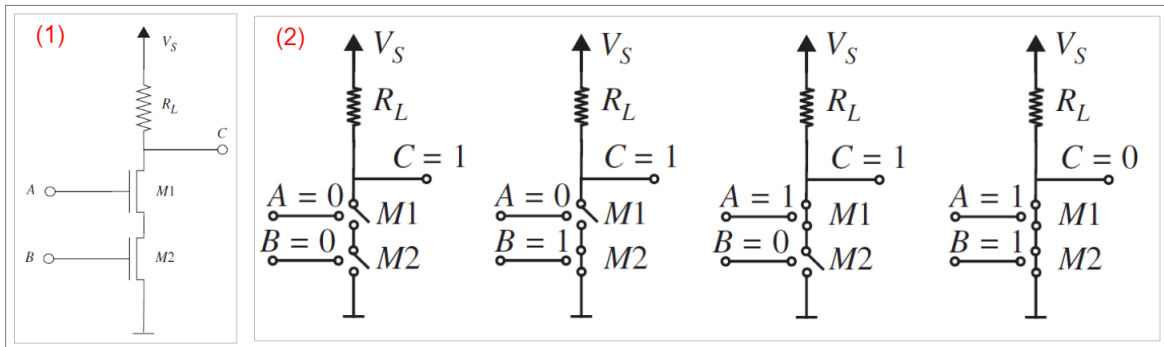


Figure 4: (1) *MOSFET NAND Gate* (2) *Equivalent Circuits for Different Inputs*

Consider the circuit shown in the figure above, which comprises a two MOSFETs and a load resistor powered by a supply voltage V_S . This circuit acts as a NAND gate. NAND gate is a universal gate that can be used to implement any Logic Function.

Logic Function Using MOSFET

We can use combinations of MOSFETs to implement various logic functions. The circuit shown in **Figure-5** implements $f = xy + z$.

I-V Characteristics of a MOSFET

The S-model of a MOSFET is a simplification of the actual I-V characteristics of a MOSFET which helps us understand the behavior of a MOSFET as a switch. But the actual I-V characteristics is very different from it.

Figure-6 represents the actual I-V characteristics of a MOSFET. There are 3 operating regions in the I-V characteristics graph: **Cut-off, Triode, Saturation**. The first two are useful if we want to use MOSFET as a switch. On the other hand, if we want to use MOSFET as an amplifier, we must operate it in the saturation region.

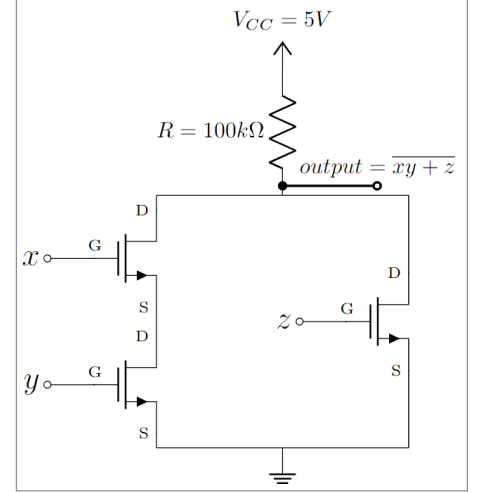


Figure 5: *Logic Function using MOSFET*

MOSFET's operating region depends on the applied voltages at Gate, Drain, Source terminals which can be seen in the I-V characteristics graph.

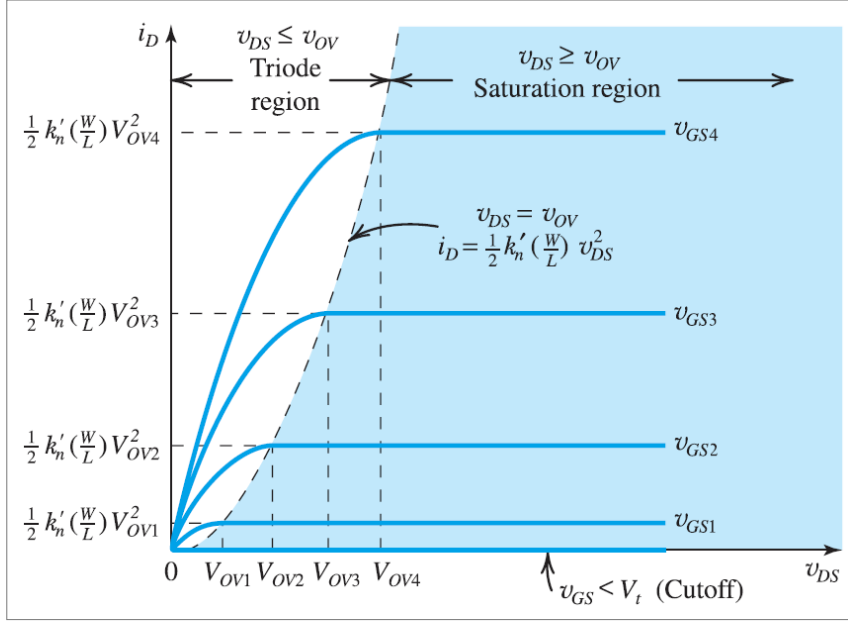


Figure 6: *I-V Characteristics of a MOSFET*

In this graph

V_T = **Threshold Voltage**

v_{OV} = **Overdrive voltage**

L = **Channel Length**

W = **Channel Width**

k'_n = **Process Transconductance Parameter**

$$v_{GS} = v_G - v_S$$

$$v_{DS} = v_D - v_S$$

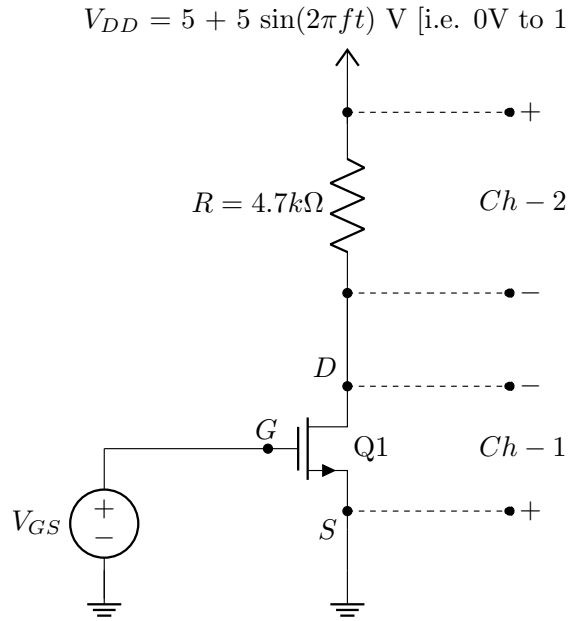
$$v_{OV} = v_{GS} - V_T$$

We can summarize the operating regions of as MOSFET as:

When	Operating Mode	Equation of Current
$v_{GS} < V_T$	Cutoff Mode	$i_D = 0$
$v_{GS} \geq V_T, v_{DS} \leq v_{OV}$	Triode Mode	$i_D = k'_n (W/L) (v_{OV} - \frac{1}{2} v_{DS}) v_{DS}$
$v_{GS} \geq V_T, v_{DS} > v_{OV}$	Saturation Mode	$i_D = \frac{1}{2} k'_n (W/L) v_{OV}^2$

The value of V_T is controlled during device fabrication and typically lies in the range of 0.3 V to 1 V. From the graph, we see that, for the same value of v_{DS} , i_{DS} increases if we increase v_{GS} . Also, for a lower value of v_{GS} , the MOSFET operates in Saturation for a lower value of v_{DS} and for higher value of v_{GS} , the MOSFET goes into Saturation for a higher value of v_{DS} .

Task-01: I-V Characteristics of a MOSFET



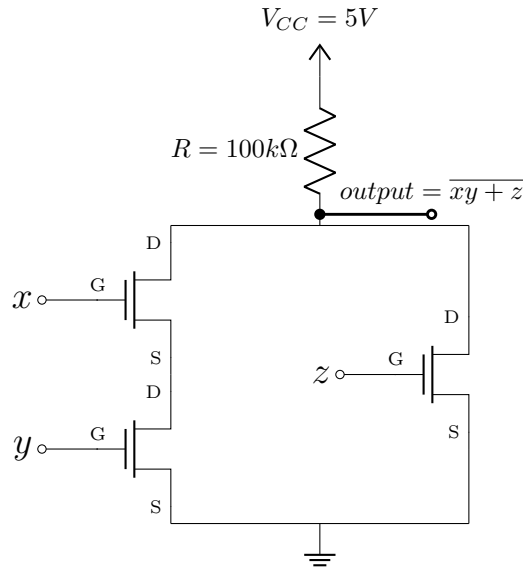
Procedure

1. Construct the circuit shown above. Use a data switch and connect it to the Gate of the MOSFET which will provide necessary voltage for V_{GS} . Use the function generator for $V_{DD} = 5 + 5 \sin(2\pi ft) \text{ V}$ [i.e. an AC voltage that varies from 0V to 10V] and set $f = 50 \text{ Hz}$.
2. Set the oscilloscope in X-Y mode. Invert the Channel-1.
3. Observe the plot in the oscilloscope when the data switch is ON and OFF. This plot shows the I-V characteristics of a MOSFET as a switch. Capture the plots using your mobile camera.
4. Now, disconnect the data switch from the gate of the MOSFET and connect the dc power supply to the gate terminal so that we can increase or decrease V_{GS} .
5. Rotate the voltage knob of the dc power supply slowly from 0V to 5V. You should observe the change in the I-V characteristics.
6. Use your mobile camera to capture the image of the I-V characteristics graphs for 3 different V_{GS} . Measure the values of V_{GS} of the captured images and write them in Data Table 1.

Data Table 1: Different Values of V_{GS} to observe the change in MOSFET I-V Characteristics

Values of V_{GS}	
1st Value	
2nd Value	
3rd Value	

Task-02: Logic Function Implementation



Circuit-1
Logic Function, $f = \overline{xy} + z$ using MOSFET

Procedure

1. On a trainer board, setup the Circuit 1.
2. Connect the gate terminals (input x, y and z) to data switches. Those switches provide 5V approximately.
3. Keeping V_{CC} constant at 5V, at first turn off the data switches connected to the gate terminal. This implies you are now applying $V_x = 0V$, $V_y = 0V$ and $V_z = 0V$. Measure the corresponding output voltage, V_{out} which should be approximately 5V which corresponds to boolean 1.
4. The boolean outputs can also be determined by the state of an LED. Connect V_{out} to one of the LEDs and check it. When the LED is ON, the boolean output is 1. Similarly, when the LED is OFF, the boolean output is 0.
5. Next, use the input voltage combinations of Data Table 2 and observe the state of LED again.
6. Verify the truth table of the Logic Function, $f = \overline{xy} + z$.

Data Table 2: Verification of the Truth Tables of Logic Function, $f = \overline{xy} + z$

Input Voltage, V_x (volt)	Input Voltage, V_y (volt)	Input Voltage, V_z (volt)	State of LED (On/Off)	Boolean Output (0 or 1)
0V	0V	0V		
0V	0V	5V		
0V	5V	0V		
0V	5V	5V		
5V	0V	0V		
5V	0V	5V		
5V	5V	0V		
5V	5V	5V		

Objective

1. To observe and understand the I-V characteristics of BJT.
2. To implement a Logic Function using BJTs and verify the truth tables.

Background Theory

Introduction to BJT

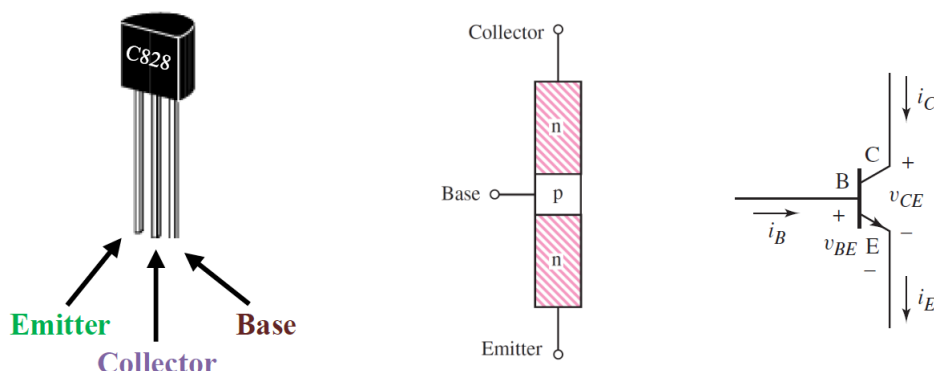


Figure 7: IC (C828), Simple Geometry and Circuit Symbol of an npn BJT

The bipolar junction transistor (BJT) is a type of transistor that is used for electrical amplification and in very-high frequency applications such as radio frequency (RF) circuits for wireless systems and high-speed switching emitter-coupled logic (ECL) gates. BJT is primarily a three terminal device consisting of the following terminals: Base (B), Emitter (E), Collector (C). There are two types of BJTs: (i) npn BJT and (ii) pnp BJT. Our discussion and experiment will be confined to npn BJT.

The figure above shows the IC, circuit symbol and simple geometry of an npn BJT. The arrowhead in the circuit symbol is always placed on the emitter terminal, and it indicates the direction of the emitter current. For an npn BJT, this direction is out of the emitter. The npn BJT contains a thin p-region between two n-regions. So the transistor consists of two pn junctions, the emitter–base junction (EBJ) and the collector–base junction (CBJ). Depending on the bias condition (forward or reverse) of each of these junctions, different modes of operation of the BJT are obtained. The operating modes are: Cut-off, Active and Saturation. The following table summarizes the modes of operation.

The active mode is the one used if the transistor is to operate as an amplifier. Switching applications (e.B. logic circuits) utilize both the cutoff mode and the saturation mode. There can be a fourth mode of a BJT called the reverse-active mode which occurs when the EBJ is reversed biased and the CBJ junction is forward biased (not shown in the following table).

BJT Modes of Operation		
Modes ↓	Emitter-Base Junction	Collector-Base Junction
Cutoff	Reverse Bias	Reverse Bias
Active	Forward Bias	Reverse Bias
Saturation	Forward Bias	Forward Bias

The active mode is the most important mode of the 3 modes of operation of BJT. Because BJT can be used as an amplifier only in this mode. BJT will be in active mode when EBJ is in Forward Bias and CBJ is in Reverse Bias. BJT operates in saturation mode when its collector current is not dependent on the base current and has

reached a maximum. This happens when both the EBJ and the CBJ are in Forward Bias. In saturation mode, huge amount of current flows through BJT and it acts like a closed switch. Cut-off mode is the opposite of saturation mode. In cut-off mode, both junctions of BJT remain reverse biased. That is why no current flows through the device (actually, very negligible amount of current flows) and the BJT acts like an open switch.

Input and Output I-V Characteristics of BJT

The I-V characteristics of a BJT depends on the circuit configuration. There are three basic configurations for connecting the BJT: the common base (CB) configuration, the common emitter (CE) configuration, and the common collector (CC) configuration. Though each configuration has their own applications, the CE is the most widely used configuration and by far the most popular for amplifiers.

In CE configuration, the emitter is the common terminal. Hence, the input is between the base and the emitter while the output is between the collector and the emitter. So, the input I-V characteristics is the variation of the base current I_B with the base-emitter voltage V_{BE} , and the output I-V characteristics is the variation of the collector current I_C with the collector-emitter voltage V_{CE} .

The following figure shows the input I-V characteristics of an npn BJT for the CE circuit configuration which illustrates the variation in I_B with respect to V_{BE} when V_{CE} is kept constant. In the graph, I_B changes exponentially as V_{BE} changes. This is obvious since the BJT's base-emitter junction is similar to a pn junction diode. So it's current voltage relationship should also be like a pn junction diode.

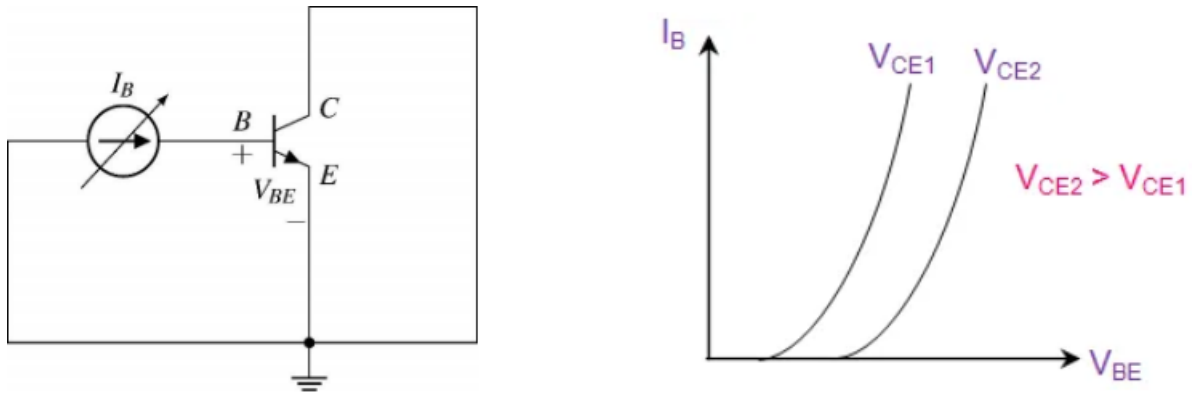


Figure 8: BJT input I-V characteristics Circuit and Graph in CE configuration

The output I-V characteristic of a BJT in CE configuration is also referred to as the collector characteristic. We are mainly interested in this one. The following circuit and graph shows the output I-V characteristics of a BJT in CE configuration. The I-V characteristics shows the variation in I_C with the changes in V_{CE} when I_B is held constant. In the graph we can see a rapid increase in collector current at the beginning. Then the collector current becomes almost constant. This graph can be divided into 3 regions:

1. Active Region (where output current becomes almost constant)
2. Saturation Region (where I_C increases rapidly)
3. Cut-off Region (where the current is zero/almost zero)

From the output I-V characteristics we see that, in the active region, if we keep V_{CE} constant, I_C increases with the increase of I_B . This relationship between I_C and I_B in active mode is actually linear in nature which can be represented by the following equation: $I_C = \beta I_B$, where β is a constant. Typically, $\beta = 50$ to 200 .

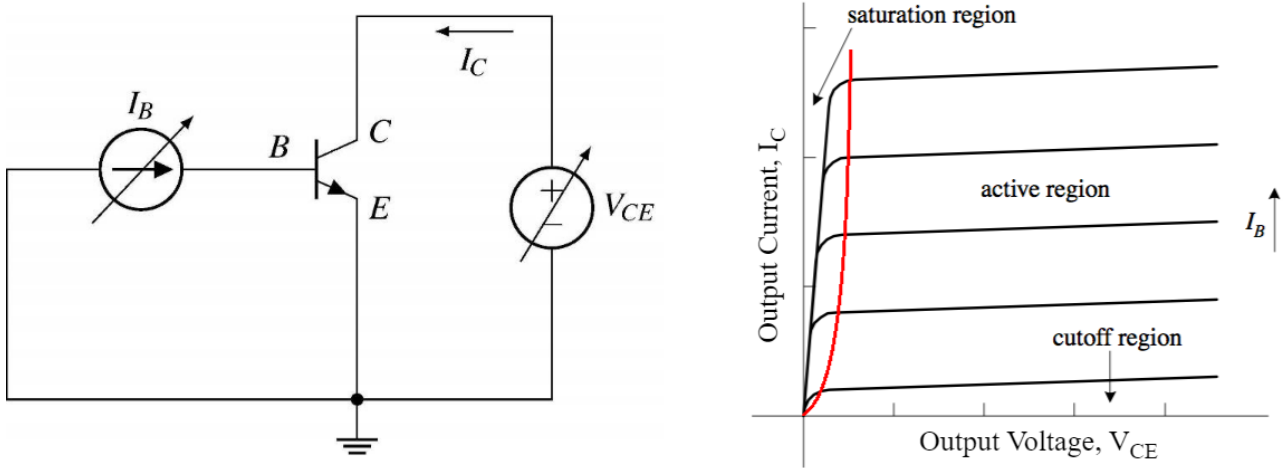


Figure 9: BJT output I-V characteristics Circuit and Graph in CE configuration

BJT as a Switch

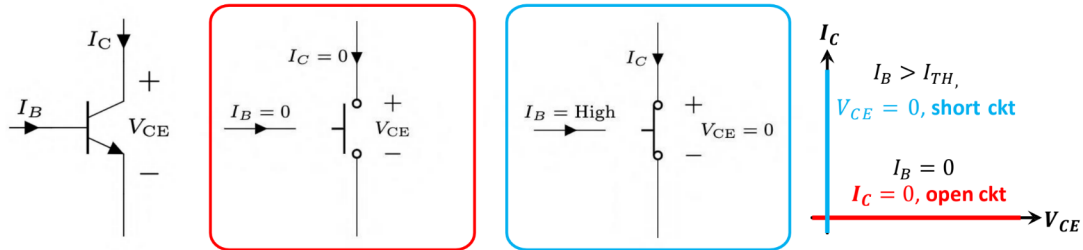


Figure 10: The S-model representation of a BJT acting as a current-controlled switch

BJT can act as a switch depending on the applied current in the input terminal. A simple circuit model known as 'Switch Model' or 'S-Model' is used to model this behavior. As seen in figure 10, when $I_B = 0$, $I_C = 0$, BJT acts like an open switch. When $I_B = \text{high}$, $V_{CE} = 0$, BJT acts like a closed switch.

Logic Gate Implementation using BJTs

BJTs can be used to construct logic gates just like MOSFETs. Unlike MOSFETs, they are current-controlled, their state is controlled by I_B .

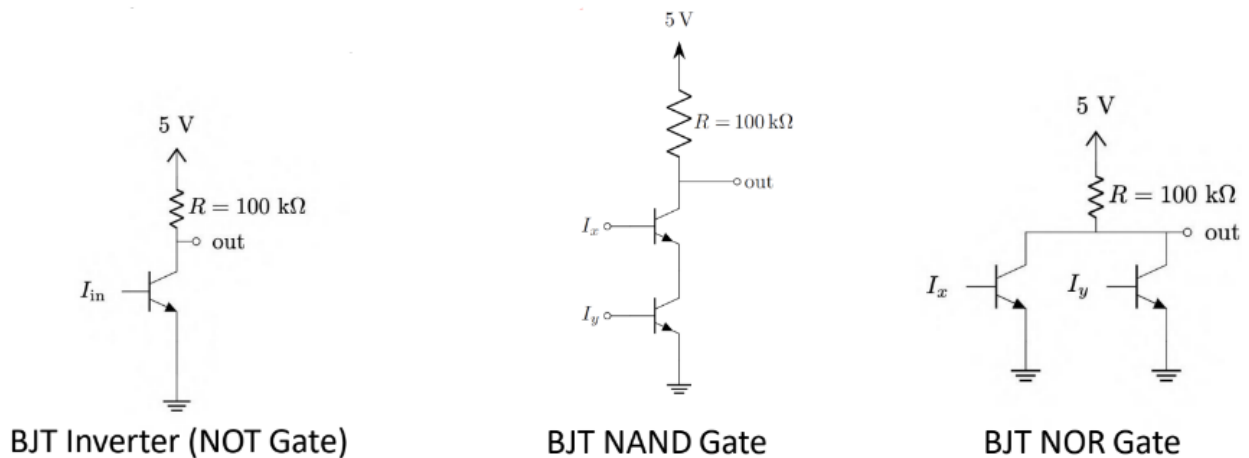
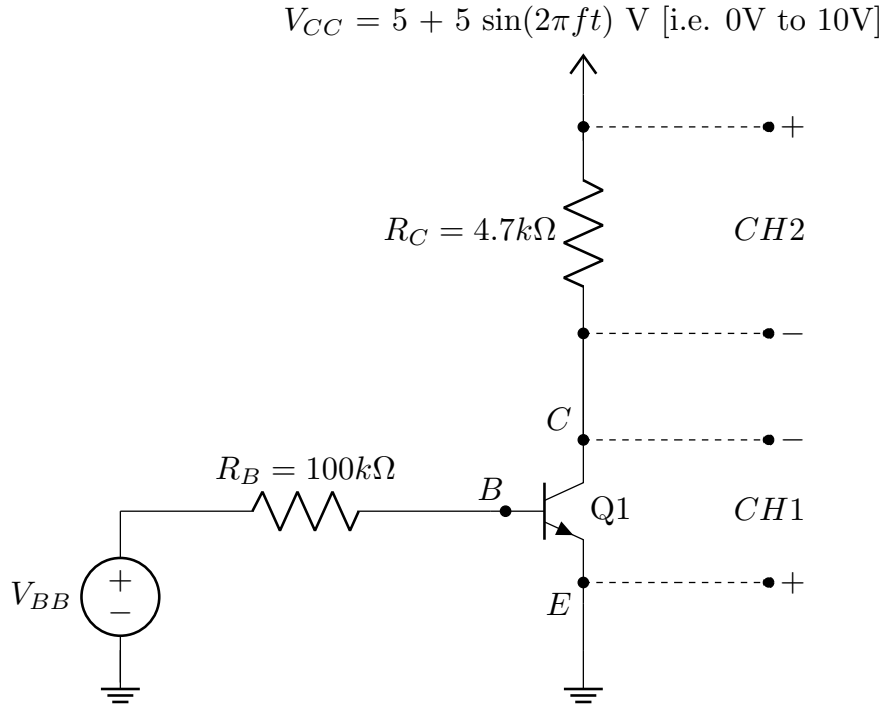


Figure 11: NOT, NAND & NOR gates using BJT

However, cascading these logic gates i.e, using the output of one logic gate as the input of another is not possible if the input is current and the output is voltage. Therefore, to translate the voltage output into a current input, we add resistors at the bases (input terminals) of the BJTs.

Task-03: I-V Characteristics of a BJT



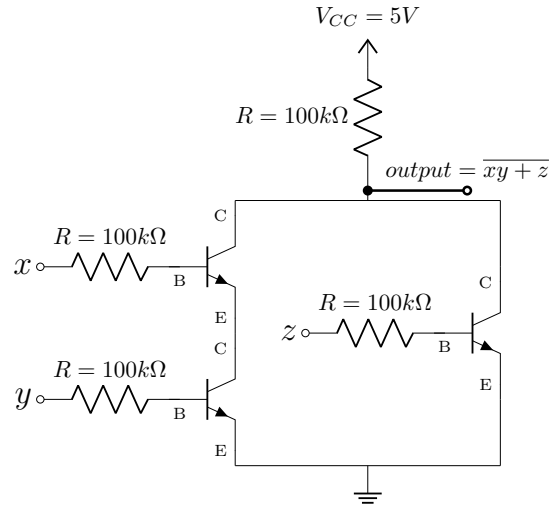
Procedure

1. Construct the circuit shown above. Use the function generator for $V_{CC} = 5 + 5 \sin(2\pi ft)$ V [i.e. an AC voltage that varies from 0V to 10V]. Set $f = 50$ Hz. Use the dc power supply for V_{BB} .
2. Invert the CH1. Set the oscilloscope in the X-Y mode.
3. Now, rotate the voltage knob of the dc power supply slowly from 0V to 5V to vary V_{BB} . You should observe the change in the I-V characteristics.
4. Use your mobile camera to capture the image of the I-V characteristics graphs for 3 different values of V_{BB} . Measure the values of V_{BB} of the captured images and write them in Data Table 3:

Data Table 3: Different Values of V_{BB} to observe the change in BJT I-V Characteristics

Values of V_{BB}	
1st Value	
2nd Value	
3rd Value	

Task-04: Logic Function Implementation



Circuit-1

Logic Function, $f = \overline{xy} + z$ using BJT

Procedure

1. On a trainer board, setup the Circuit 1.
2. Connect the gate terminals (input x and y) to data switches. Those switches provide 5V approximately.
3. Keeping V_{CC} constant at 5V, at first turn off the data switches connected to the gate terminal. This implies you are now applying $V_x = 0V$, $V_y = 0V$ and $V_z = 0V$. Measure the corresponding output voltage, V_{out} which should be approximately 5V which corresponds to boolean 1.
4. The boolean outputs can also be determined by the state of an LED. Connect V_{out} to one of the LEDs and check it. When the LED is ON, the boolean output is 1. Similarly, when the LED is OFF, the boolean output is 0.
5. Next, use the input voltage combinations of Data Table 4 and observe the state of LED again.
6. Verify the truth table of the Logic Function, $f = \overline{xy} + z$.

Data Table 4: Verification of the Truth Tables of Logic Function, $f = \overline{xy} + z$

Input Voltage, V_x (volt)	Input Voltage, V_y (volt)	Input Voltage, V_z (volt)	State of LED (On/Off)	Boolean Output (0 or 1)
0V	0V	0V		
0V	0V	5V		
0V	5V	0V		
0V	5V	5V		
5V	0V	0V		
5V	0V	5V		
5V	5V	0V		
5V	5V	5V		

Task-05: Report

1. Cover page [include course code, course title, name, student ID, group, semester, date of performance, date of submission]
2. Attach the signed Data Sheet.
3. Attach the captured photos of all the waveforms and measurements you have observed in the Oscilloscope. Each photo should contain necessary description.
4. Answer the questions of the Test Your Understanding section.
5. Add a brief Discussion at the end of the report. For the Discussion part of the lab report, you should include the answers of the following questions in your own words:
 - What did you learn from this experiment?
 - What challenges did you face and how did you overcome the challenges? (if any)
 - What mistakes did you make and how did you correct the mistakes? (if any)
 - How will this experiment help you in future experiments of this course?

Signature of the lab faculty

Test Your Understanding

Answer the following questions:

1. We can use the MOSFET as a switch. Which operating regions do we need for this purpose and why?
Answer:

2. Draw the VTC (V_{DS} vs V_{GS}) graph of a MOSFET operating as a switch.

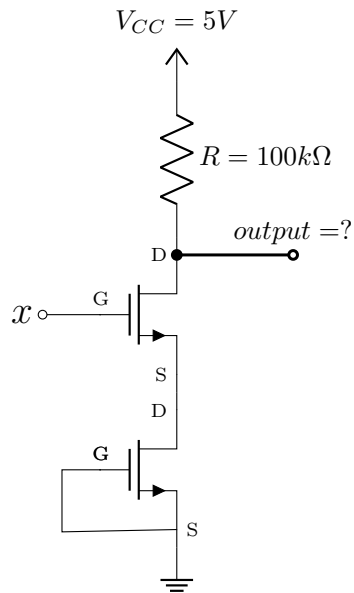
Answer:

3. Draw a circuit using MOSFETs that implements the following logic function, $f = (x+y)z$

Answer:

4. When a MOSFET is in triode mode, does it display zero, finite or infinite resistance? Justify your answer.

Answer:



5. What is the value of the output of the circuit above?

Answer:

6. If $V_{DS} = 3V$, $V_{GS} = 3V$ and $V_T = 1V$, find the gate current, drain current and source current of a MOSFET given that $k = 0.5mA/V^2$.

Answer:

7. We can use the BJT as a switch. Which operating regions do we need for this purpose and why?

Answer:

8. What's the primary difference between a MOSFET and a BJT when they are both acting as switches?

Answer:

9. Suppose we want to cascade two BJT logic gates (use the output of the first logic gate as one of the inputs of the second one). Why do we need to add a resistor at the input of the second logic gate?

Answer:

10. Draw a circuit using BJTs that implements the following logic function, $f = (\overline{x + y}).z$

Answer:

11. Suppose, a BJT is operating in the active region where, $I_B = 0.01mA$. If $\beta = 100$, find the value of I_C and I_E .

Answer: